



Aakash
Medical | IIT-JEE | Foundations

Code A Phase 1

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MM : 45

Subjective Test_OYMP1-2526_PST-01A

Time : 90 Min.

Topics Covered:

Physics: Electric Charges & Fields, Electrostatic Potential and Capacitance

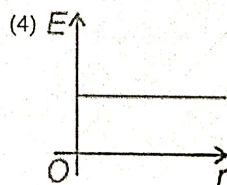
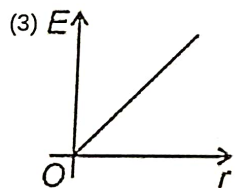
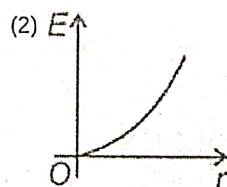
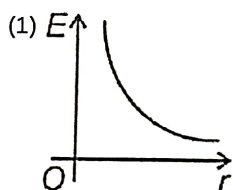
Chemistry: Solutions

Biology: Sexual Reproduction in Flowering Plants, Human Reproduction

PHYSICS

Q1. The variation of electric field E with distance r from an infinite long uniformly charged straight wire is

[1]



Q2. The SI unit of electric field is _____

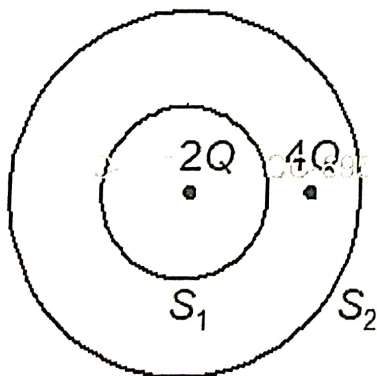
[1]

Q3. State Kirchhoff's rule of current distribution in an electrical network.

[1]

Q4. Consider two hollow concentric spheres S_1 and S_2 enclosing charges $2Q$ and $4Q$ respectively. Calculate the ratio of electric flux through them.

[2]

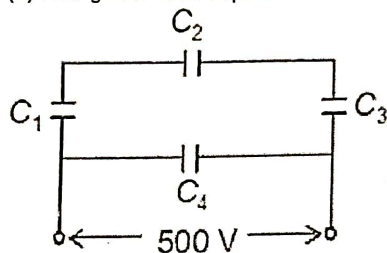


Q5. Determine the electrostatic potential energy of the system consisting of two charges $7 \mu\text{C}$ and $-2 \mu\text{C}$ placed at $(-9 \text{ cm}, 0, 0)$ and $(9 \text{ cm}, 0, 0)$. Also find the work required to separate the two charges infinitely away from each other.

[2]

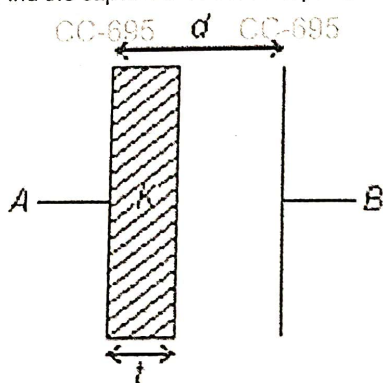
Q6. A network of four capacitors each of capacitance $12 \mu\text{F}$ is connected to a 500 V supply as shown in the figure. Determine [3]

- (a) Equivalent capacitance of the network and
(b) Charge on each capacitor



OR

Find the capacitance between point A and B of parallel plate capacitor as shown in figure. [3]



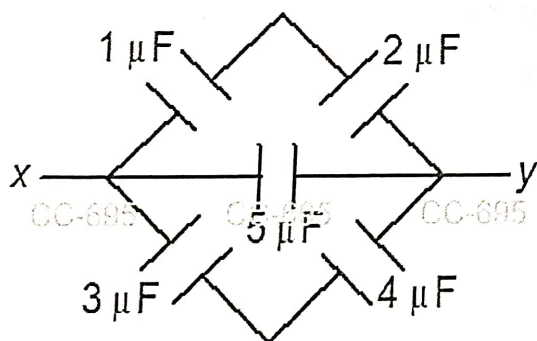
Q7. (a) A capacitor of capacitance $10 \mu\text{F}$ is charged by connecting a battery of negligible internal resistance and emf of 20 V across it. [1]
Calculate the amount of work done by the battery in charging the capacitor fully?

(b) Write the two basic properties of charge. [2]

(c) Two charges of a dipole $-4 \mu\text{C}$ and $+4 \mu\text{C}$ are placed at the points $A(1, 0, 4) \text{ m}$ and $B(2, -1, 5) \text{ m}$ located in an electric field $\vec{E} = 0.20\hat{i} \text{ V/cm}$. Calculate the torque acting on the dipole. [2]

OR

(a) Five capacitors are connected as shown in the diagram. Calculate equivalent capacitance between the points x and y. [1]



(b) Explain the underlying principle of working of a parallel plate capacitor. If two similar plates, each of area A having surface charge density $+\sigma$ and $-\sigma$ are separated by d in air. Expression for the capacitance of the capacitor so formed. [2]

(c) Two point charges of $+1 \mu\text{C}$ and $+4 \mu\text{C}$ respectively are placed at 12 cm apart. Find the position of the point where electric field is zero. [2]

Q8. Define isotonic solutions. [1]

Q9. Which of the following is not a colligative property? [1]

- (1) Vapour pressure (2) Depression of freezing point
(3) Osmotic pressure (4) Elevation of boiling point

Q10. Calculate the molarity of an aqueous solution in which 30% ($\frac{w}{v}$) urea is present. [2]

Q11. A solution of non-electrolyte solute causes depression in freezing point of solvent by 0.40 K. The freezing point depression constant of the given solvent is $5.12 \text{ K kg mol}^{-1}$. Find the molality of solution. [2]

Q12(a) Calculate the osmotic pressure of an aqueous solution of $0.1 \text{ M Na}_3\text{PO}_4$ (ionization to 60%) at 27°C . [2]

(b) Write two differences between ideal solutions and non-ideal solutions. [2]

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OR

(a) Calculate the boiling point of an aqueous solution of 0.1 m NaCl (99% ionisation) ($K_b(\text{H}_2\text{O}) = 0.52 \text{ K molal}^{-1}$) [2]

(b) Why constant boiling mixtures behave like one single component? [2]

Q13(a) What is meant by negative deviation from Raoult's law? Give an example. [2]

(b) Arrange the following aq. solutions in increasing order of their freezing points [3]

- (i) 0.1 M KCN
(ii) $0.2 \text{ M Na}_2\text{SO}_4$
(iii) 0.3 M urea

OR

(a) State Henry's law and mention one of its important application. [2]

(b) The vapour pressure of pure liquids A and B are 450 and 700 mm of Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg. Also find the composition of vapour phase [3]

BIOLOGY

Q14. A gynoecium of *Michelia* is [1]

- (1) Multicarpellary and apocarpous (2) Bicarpellary and syncarpous
(3) Multicarpellary and syncarpous (4) Bicarpellary and apocarpous

OR

(22 + Y) is the genetic constitution observed in [1]

- (1) Spermatogonia (2) Primary spermatocytes
(3) Secondary spermatocytes (4) Primary oocytes

Q15. If the parent produces unisexual flowers then there is no need of emasculation. Why? [2]

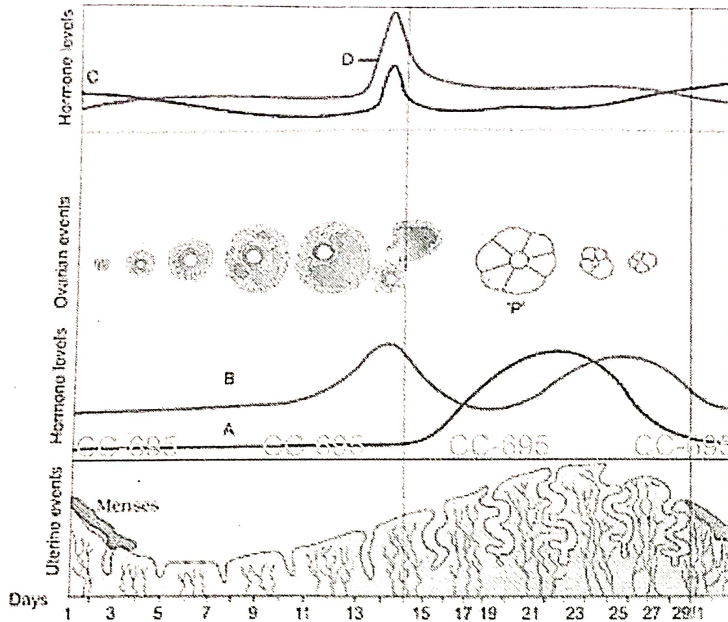
Q16. Why do doctors recommend breast feeding during the initial period of infant growth? [2]

Q17. Differentiate between microsporogenesis and megasporogenesis [2]

OR

- (a) Write the function of oxytocin. [2]
(b) Write a short note on foetal ejection reflex.

Q18. What is the importance of apomixis in hybrid seed industry? How does apomixis mimic sexual reproduction? [3]



[3]

With reference to the above schematic representation of various events during a menstrual cycle, answer the following questions.

- Name the hormones labelled as A, B, C and D.
- What does 'P' represent? Name the hormone it secretes along with its function.

Q19(a) Draw a well labelled diagram of a mature embryo sac and label any six parts.

[3]

- Explain how geitonogamy is functionally similar to cross-pollination and genetically similar to autogamy.

[2]

OR

- Draw a well labelled diagram of the male reproductive system of humans.
- State the differences between meiosis-I and meiosis-II of spermatogenesis and oogenesis.

[5]